

Xuhui Lu

Ph.D. Candidate in AI for Software Engineering · Reliable Software Systems
Beihang University · Expected Jun. 2027
Beijing, China | luxuhui@buaa.edu.cn



Research Interests

- Reliable software engineering and AI agents for software development
- Repository-level software comprehension, function-call graphs, and retrieval-augmented generation
- Program analysis, automated repair and testing, specification discovery, and software change validation

Education

Ph.D. Program / Ph.D. Candidate

Beihang University

Sep. 2021 - Expected Jun. 2027

Beijing, China

- Advisor: Prof. Zheng Zheng

B.Eng. in Automation

Beihang University

Sep. 2017 - Jun. 2021

Beijing, China

Publications

Peer-Reviewed Journal Articles

1. **Lu, X.**, Zheng, Z., Qin, F., Ma, X., and Cai, Q. (2025) “Functional and Defect Study in Deep Learning Libraries: A Complex Network Perspective,” *IEEE Transactions on Reliability*, **74(4)**, 4900–4914.
2. Ding, W., Liu, Z., **Lu, X.**, Du, X., and Zheng, Z. (2025) “KPAMA: A Kubernetes based tool for Mitigating ML system Aging,” *Journal of Systems and Software*, **226**, 112389.
3. Wan, X., Zheng, Z., Qin, F., and **Lu, X.** (2024) “Data Complexity: A New Perspective for Analyzing the Difficulty of Defect Prediction Tasks,” *ACM Transactions on Software Engineering and Methodology*, **33(6)**, Article 141, 1–45.
4. Wan, X., Zheng, Z., Qin, F., **Lu, X.**, and Qiu, K. (2024) “Adjusted Trust Score: A Novel Approach for Estimating the Trustworthiness of Software Defect Prediction Models,” *IEEE Transactions on Reliability*, **73(4)**, 1877–1891.

Manuscripts Under Review

1. **Lu, X.**, Fan, Y., Tang, Y., Ma, X., Wan, X., Qin, F., Yuan, W., and Zheng, Z. (2026) “FCG-RAG: Structure-Aware Retrieval-Augmented Generation for Software Comprehension,” *ACM Transactions on Software Engineering and Methodology*. **Under review.**
2. Ma, X., Wang, Z., **Lu, X.**, Du, X., Li, C., Li, J., Li, H., and Zheng, Z. (2026) “DeepGroundRepair: From API-Grounded Generation to Shape-Aware Repair for Deep Learning Programs,” *ACM Transactions on Software Engineering and Methodology*. **Under review.**

Research & Industry Experience

School–Enterprise Joint Research

Wuhan Guide Infrared Co., Ltd.

Sep. 2023 - Sep. 2025

Wuhan, China

- Conducted research on object detection and open-vocabulary object detection for infrared imagery, with emphasis on generalization to previously unseen categories and practical deployment.

Research Intern, Trusted Theory, Technology and Engineering Lab

Huawei 2012 Laboratories

Jul. 2023 - Sep. 2023

China

- Participated in industry research on trustworthy software and engineering.

Reliability Testing for AI Systems

Beihang University–Huawei Joint Project

May 2021 - Aug. 2022

Beijing, China

- Performed static analysis of the MindSpore framework, constructed function-call graphs, mapped defects to graph elements, and analyzed network-topology characteristics for software reliability studies.

Honors & Research Funding

Provincial Research Funding

Hubei Provincial Engineering Master’s/Doctoral School–Enterprise Joint Training Practice Project; RMB 200,000; primary student contributor (first-listed), China

2024

National Excellence Engineer Scholarship

Beihang University, Beijing

2023

First-Class Graduate Academic Scholarship

Beihang University, Beijing

2022

Teaching Experience

Teaching Assistant

Fundamentals of Software Technology (2022)

Applied Discrete Mathematics (2021)

2021–2022

Beihang University

Academic Service

Reviewer

IEEE Transactions on Industrial Informatics; IEEE International Conference on Software Quality, Reliability and Security (QRS)